

Name: _____

Class: _____

ACTIVITY SHEET

Chapter 5 Revision

1 Match each term with the appropriate description using an arrow.

a	anions
b	atomic mass unit
c	species
d	Avogadro constant (N_A)
e	balanced equation
f	cations
g	chemical equation
h	chemical formula
i	empirical formula
j	formula unit
k	ionic compound
l	lattice
m	mass (m)
n	molar mass (M)
o	mole of a substance
p	molecular compound
q	molecular formula
r	molecule
s	percentage composition
t	products
u	reactants
v	relative atomic mass (A_r)
w	relative formula mass (M_r)
x	relative molecular mass (M_r)
y	stoichiometry

A	amount of substance containing 6.02×10^{23} particles
B	compound consisting of oppositely charged ions held in a lattice
C	consists of different types of atoms covalently bonded together
D	atoms, ions or molecules involved in a chemical reaction
E	empirical formula of any ionic compound or network solid
F	formula of an element or compound
G	gives actual number of different atoms present in a molecule
H	group of atoms held together by covalent bonds
I	mass of an atom compared to the mass of a carbon-12 atom
J	mass of one mole of a substance
K	mass of a molecule compared to the mass of a carbon-12 atom
L	negatively-charged ions
M	new substances formed in a chemical reaction
N	number of atoms in exactly 1 g of the carbon-12 isotope
O	numbers of each type of atom are the same on both sides
P	percentage by mass of each element found in a compound
Q	positively-charged ions
R	quantity of matter in a substance
S	1/12 the mass of one carbon-12 atom
T	regular arrangement of particles within a solid
U	mass of a formula unit compared to the mass of a C-12 atom
V	simplest whole number ratio of atoms present in a compound
W	starting substances in a chemical reaction
X	study of the amounts of substances in a chemical reaction
Y	uses chemical symbols for reactants and products in a reaction

2 Explain the differences between the terms in each of the items below.

a Relative atomic mass, relative molecular mass and relative formula mass

b Empirical formula and molecular formula

c A mole and the molar mass of a substance

3 Decide whether each of the following statements is true or false. If the statement is false, correct it.

a The relative atomic mass of an atom is its mass relative to any carbon atom. (True / False)

b The relative molecular mass of a molecule is the sum of the relative atomic masses of the atoms in that molecule. (True / False)

c All pure substances exist as molecules. (True / False)

d The formula of an ionic compound gives the ratio of the atoms present in the compound. (True / False)

e The Avogadro constant is equal to the number of atoms in exactly 12 g of the carbon-12 isotope. (True / False)

f One mole of a substance would contain 6.02×10^{23} particles of that substance. (True / False)

g The formula of any ionic substance is always an empirical formula. (True / False)

h Molar mass is an actual mass that is measured in moles. (True / False)

i A molecular formula gives the exact number of the different kinds of atoms in the molecule. (True / False)

j Stoichiometry focuses on the actual amounts of reactants and products in a reaction. (True / False)

4 Complete the following table by adding the symbol, unit and unit symbol for each quantity.

Quantity	Symbol	Units	Unit symbols
Relative atomic mass		atomic mass units	amu
Relative molecular mass			
Relative formula mass			
Mass			
Number of moles			
Molar mass			

- 5 State the three steps involved in working out the relative molecular mass of a molecule, given that you know the relative atomic masses of the atoms involved.

Step 1:

Step 2:

Step 3:

- 6 State the three steps involved in calculating the relative formula mass of an ionic compound, given that you know the relative atomic masses of the atoms involved.

Step 1:

Step 2:

Step 3:

- 7 State the three steps involved in determining the percentage composition by mass of each element in a compound.

Step 1:

Step 2:

Step 3:

- 8 State the single step involved in finding how many moles there are in a specified number of atoms, molecules or formula units.
- 9 State the single step involved in working out how many particles there are in a specified number of moles of atoms, ions or molecules.
- 10 State the single step involved in calculating the molar mass of an element or compound, given its relative molecular or formula mass.
- 11 State the three steps involved in finding how many moles of an element or compound there are in a given mass of it.
- Step 1:
- Step 2:
- Step 3:

12 State the three steps involved in determining the mass of a given number of moles of an element or compound.

Step 1:

Step 2:

Step 3:

13 State the two steps involved in finding the number of moles of a reactant or product when the number of moles of another reactant or product is specified.

Step 1:

Step 2: